Literature Search and Selection Record

Supplementary methodological documentation supporting Section 2.2, “Literature Search and Selection Approach”.

1. Scope and search sources

The literature review adopted a focused thematic approach to examine large-language-model reliability, Retrieval-Augmented Generation (RAG), educational applications of RAG, programming assistance and transparency in educational AI. Searches were conducted using Google Scholar and Semantic Scholar. The identification and screening of publications followed the research themes established for CurriculumAI, with an emphasis on directly relevant peer-reviewed studies.

Theme	Representative search terms	Purpose in the review
LLM reliability	“LLM hallucination”; “hallucination taxonomy”	To identify research on hallucination, factuality and evidential reliability.
RAG architecture	“retrieval-augmented generation”; “RAG failure modes”	To identify research on retrieval-based grounding, attribution and failure conditions.
Educational RAG	“RAG chatbot education”; “educational chatbot”	To identify studies applying retrieved-context support in educational settings.
Programming education	“ChatGPT programming education”; “student trust AI programming”	To identify research on programming assistance, curricular fit and learner perceptions.
Research methodology	“Design Science Research”; “DSRM information system”	To support the methodological framing of the artefact-centred study.

The terms shown are those retained in Section 2.2 of the dissertation. Other related combinations were used during literature exploration.

2. Screening and selection process

26	9	17
Initially shortlisted papers	Excluded after screening	Studies retained for Chapter 2

An initial set of 26 papers was shortlisted around the five review themes. The shortlist was screened for direct relevance to CurriculumAI’s research problem, peer-reviewed publication status and a preference for research published from 2020 onwards. Publications outside the study’s immediate scope, arXiv-only preprints and older works outside the preferred review period were not retained in the Chapter 2 synthesis. A total of nine papers were excluded at screening, leaving 17 studies for the thematic review.

The retained publications were organised by their contribution to reliability, RAG design, educational use, programming assistance or transparency. Foundational methodological references, including Hevner et al. (2004) and Peffers et al. (2007), informed Chapter 3 separately and were not included in the 17 studies examined in Chapter 2.

3. Studies retained for the thematic review

The 17 studies retained for Chapter 2 are listed below according to their principal analytical role in the review.

Analytical strand	Study	Role in the review	Section
LLM reliability	Bender et al. (2021)	Fluency and limitations of language modelling	2.4
	Ji et al. (2023)	Hallucination, factuality and source faithfulness	2.3–2.4
	Sun et al. (2024)	Types of generated error and taxonomy limitations	2.4
RAG mechanisms	Lewis et al. (2020)	Foundational retrieval-augmented generation	2.5
	Barnett et al. (2024)	Failure points and absent versus missed evidence	2.5
	Klesel and Wittmann (2025)	RAG architecture and system limitations	2.5
	Byun et al. (2024)	Retrieved-context support and response grounding	2.5
	Gao et al. (2023)	Citation-to-claim support considerations	2.5
Educational RAG	Lang and Gürpinar (2025)	Course-based support and student perceptions	2.6
	Thüs et al. (2024)	Educational RAG and student engagement with literature	2.6
	Wijaya and Purwarianti (2024)	Programming tutoring and retrieval-based QA	2.6
Programming AI	Kazemitabaar et al. (2024)	Programming-assistant use in classroom settings	2.7
	Penney et al. (2025)	Help-seeking behaviour and reliability concerns	2.7
	Haindl and Weinberger (2024)	Student experiences in programming courses	2.7
	Najimi et al. (2026)	Comparative evaluation in programming education	2.7
AI transparency	Yan et al. (2024)	Disclosure and ethical issues in educational AI	2.8
	Bond et al. (2024)	Ethics, rigour and higher-education AI research	2.8

4. Connection to the research gap

The thematic synthesis in Section 2.9 connected the reviewed evidence to the bounded research gap in Section 2.10. Studies on hallucination and RAG established why factual correctness, retrieved-context support and source attribution require separate scrutiny. Research on programming assistants and educational RAG highlighted curriculum alignment, the limits of course coverage and the difference between measured answer quality and student perceptions. These distinctions informed CurriculumAI’s source-transparent two-tier design and the separate technical and student evaluation measures.